

Create a 30-day learning roadmap. Include: - Weekly milestones - Daily tasks - Resources - Projects - Final outcome Make it practical and beginner-friendly.

🌀 Thought for 7 seconds >

Assumption: 30-Day Python Programming Roadmap (Beginner-Friendly)

No prior coding experience needed. Expect 1–2 hours of practice daily.

Week 1: Python Basics

Milestone: Write simple scripts with variables, conditionals, and loops.

Day	Tasks	Resources
1	Install Python & VS Code; print() and input()	Python.org , Automate the Boring Stuff – Ch 1
2	Variables, data types (int, float, str, bool)	W3Schools Python Variables
3	String methods & f-strings	Real Python “String Formatting”
4	if/elif/else statements	YouTube: Corey Schafer – Conditionals
5	for loops and range()	freeCodeCamp “Python Loops”
6	while loops & break/continue	Practice with small number-guessing game
7	Mini project: “Rock, Paper, Scissors” game	Replit / Google Colab

Week 2: Data Structures & Functions

Milestone: Store and manipulate collections of data, write reusable functions.

Day	Tasks	Resources
8	Lists (append, remove, slicing)	W3Schools Python Lists
9	Tuples & sets	Real Python "Lists vs Tuples"
10	Dictionaries (keys, values, items())	freeCodeCamp – Python Dictionaries
11	Define functions, parameters, return	Corey Schafer – Functions
12	Scope and default arguments	Python.org "Function tutorial"
13	Lambda functions & list comprehensions	YouTube: Programming with Mosh
14	Mini project: To-Do List CLI app	(Add tasks, mark done, view list)

Week 3: Files, Errors & Modules

Milestone: Read/write files, handle errors, use built-in modules.

Day	Tasks	Resources
15	File I/O: open(), read(), write()	Real Python "Reading and Writing Files"
16	Working with CSV files (csv module)	GeeksforGeeks "Python CSV"
17	Try/except/else/finally for errors	freeCodeCamp – Exception Handling
18	Importing modules & creating your own	Python.org "Modules" tutorial
19	Useful built-ins: random, datetime, math	W3Schools Python Modules

Day	Tasks	Resources
20	Basic debugging (print, pdb)	YouTube: Tech With Tim – Debugging
21	Mini project: Habit tracker (log daily habits to a CSV file, show stats)	Google Colab

Week 4: Mini Library & Final Capstone

Milestone: Combine everything into a polished project.

Day	Tasks	Resources
22	Introduction to pip & external packages	Real Python “Using pip”
23	Install and try requests (fetch a public API)	JSONPlaceholder guide
24	Install and try pandas (basic DataFrame)	Kaggle “Pandas course” (first two lessons)
25	Plan your final project: e.g., Expense Tracker, Quiz App, Weather CLI	Think of a problem you care about
26	Code skeleton + file structure	No new resource – start coding
27	Implement core features (error-handled user input, file storage)	Revisit previous days as needed
28	Add a simple data visualisation (matplotlib) for expense tracker	Matplotlib “Quick start guide”
29	Testing & polishing	Ask a friend to try your script

Day	Tasks	Resources
30	Final outcome: Upload project to GitHub with a README	GitHub Desktop tutorial

Final Outcome

- **Portfolio of 3 mini projects** (Rock-Paper-Scissors, To-Do List CLI, Habit Tracker)
- **One capstone project** of your choice (e.g., Expense Tracker with graphs, or a CLI Quiz that loads questions from a CSV)
- **Basic GitHub repository** demonstrating your code and documentation
- **Ability to write 50-100 line Python scripts** that handle user input, files, errors, and external libraries

Practical Tips for Success

- Code every single day – even 30 minutes is enough.
- Use **Google Colab** (no install) or **Replit** for quick experiments.
- Stuck? Use **Python Tutor** to visualise execution or ask ChatGPT to explain a concept.
- Don't memorise syntax – keep W3Schools or Python docs open.
- Join **r/learnpython** or freeCodeCamp Discord for community help.

Would you like a version for **web development**, **data analysis**, or another topic instead?
Just ask!